The real increment test

the ocean absorbs 99% of it — but PISM is collapsing

2026-07-13, evening. Follows `dynamics_seam_2026-07-13`.

Abstract

The previous note established that the CORE3(observed)→PISM cavity swap is a one-off monster that production never has to survive, and that cold-starting the ocean on PISM's cavity sidesteps it. That was done, chunk 1 ran a clean model year, PISM ran its 10 years, and for the first time the remap was asked to carry *a year of genuine spun-up ocean dynamics across a real PISM increment*. It failed — but the way it failed inverts the conclusion. **The ocean absorbed ∼99% of the change** (CFLz *peaked and then decayed*; η had zero NaN; only **17 of 209 172 nodes** misbehaved). What is actually wrong is upstream: **PISM is in massive disequilibrium and is shedding its ice shelves**. The Ross front retreats ≈4° of latitude (≈440 km) in a *single* 10-yr leg, and **1399 nodes have their ice removed outright**, ringing the whole Antarctic margin. The "gentle 22 m increment" framing used to justify this whole strategy was wrong.

1 What the increment test actually did

Chunk-3 (awiesm3 1901) crashed at **day 15.66** (η_n NaN check). But read the diagnostics before concluding anything — they look nothing like the monster swap:

	monster swap (CORE3→PISM)	**real increment** (this leg)
CFLz over time	4.46 → 6.43 → 7.25 → 9.24	5.26 → **5.78** → 5.40 → 4.51 → **4.15**
	monotonic runaway	**peaks, then DECAYS**
died at	day ∼4	day 15.7
η_n NaN	yes	**none** — all 209 172 nodes finite
$\lvert\eta\rvert > 3$ nodes	3611, basin-wide, radiating	**17**, one tight cluster
	mostly *unchanged* nodes	**all 17 on changed nodes**

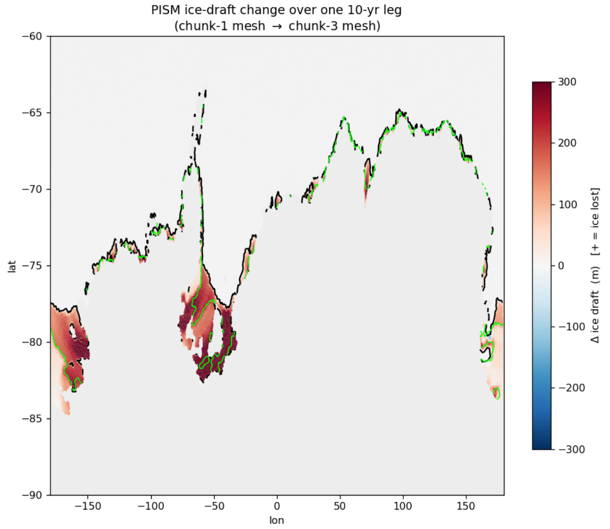
The CFLz *decays*: the ocean is *absorbing* the geometry change, not running away from it. And the 17 offending nodes share one signature exactly:

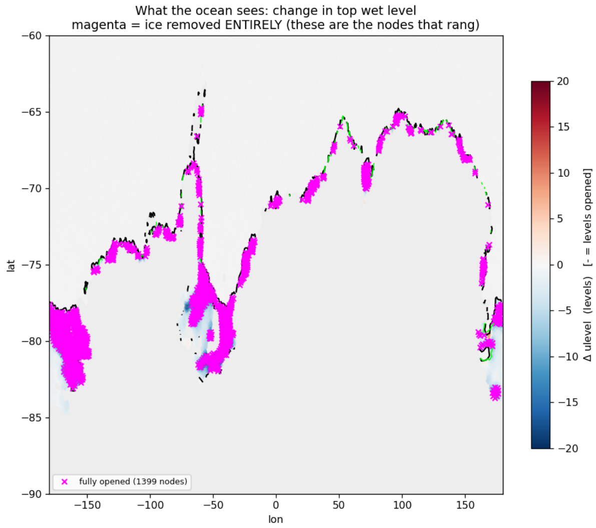
all 17 ringing nodes	`ulev 17` → `1`
i.e.	16 ocean levels opened *at once*
η values	+24.4, −17.9, +15.9, +8.3, +5.3
	alternating sign ⇒ grid-scale **checkerboard**

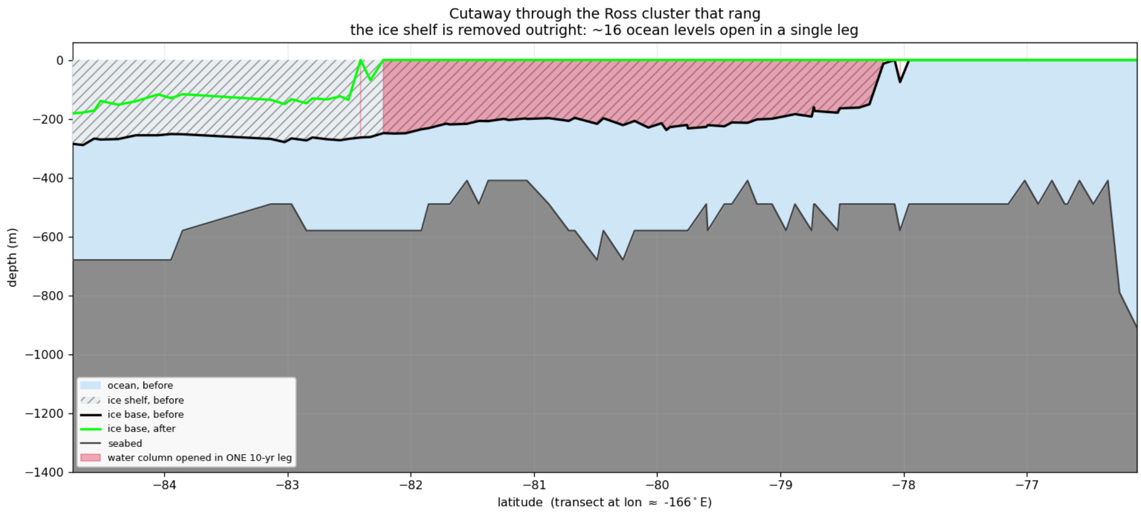
These are nodes where the ice shelf above them was **removed entirely** in one leg. They get donor water, zero velocity, and a free surface that cannot match their still-cavity neighbours ($\eta \equiv 0$ under ice) — and they ring. It is the same seam physics as before, but now confined to the *extreme tail* of the change rather than smeared over thousands of nodes.

2 The finding: PISM is collapsing

Why is the tail so extreme? Because the increment is not gentle at all.







The cutaway is the clearest statement of the problem: along this transect the ice base goes from $-200 \ldots -280$ m to **zero (open ocean)** across the whole span $-82.2°$ to $-78.2°$ latitude. **The Ross ice-shelf front retreats ≈4° of latitude (≈440 km) in ONE 10-yr leg.** The plan view shows this is not a Ross anomaly: the 1399 fully-opened nodes (magenta) ring the *entire* Antarctic margin.

changed nodes this leg	3248	(1.55% of mesh)
$\lvert\Delta\text{ulev}\rvert \geq 10$		1582
nodes with the ice removed ENTIRELY		**1399**
mean $\lvert\Delta\text{draft}\rvert$	**2.9 m**	← *meaningless*
max $\lvert\Delta\text{draft}\rvert$		**923 m**
PISM floating cells	24 815 → 20 093	(**−24% in 10 yr**)

The mean is a lie. It is diluted by the vast unchanged interior; the distribution is wildly skewed. The earlier claim that a 10-yr PISM increment is a gentle ≈22 m change — which is what motivated the entire "sidestep the swap and the increments will be fine" strategy — was **wrong**, and wrong in a way that mattered.

PISM, started from the inherited spin-up state, is **not in equilibrium**: it is shedding ice shelves catastrophically. No ocean model should be expected to swallow 440 km of ice-front retreat in one coupling step.

3 The encouraging half

Of the **1399** nodes whose ice was removed outright, only **17 rang**. The ocean absorbed ∼99% of even *this*, with the vertical-CFL transient peaking and then decaying. The remap machinery — which we have now shown to be bit-exact on unchanged nodes, inversion-free, geometrically sound, and (after the fix below) free of manufactured values — is performing far better than a "crashed at day 15" headline suggests.

4 A real remap bug fixed on the way

The first attempt at this leg died at **mstep 1** with `found temperture becomes NaN or <-5.0, >60`. The remapped restart contained −45.02°C — a value that exists *nowhere* in the source (wet range $[-2.32, 41.72]$, dry cells all 0.0). The remap *manufactured* it. The column made the mechanism obvious:

$$1.22,\ -0.99,\ -3.19,\ -5.39,\ -8.69,\ -13.09,\ -18.6,\ -25.2,\ -34.01,\ \mathbf{-45.02}$$

Smooth and accelerating: an **unbounded linear extrapolation**. For columns that get *deeper* across a mesh change, the remap fitted a straight line through the last two old levels and continued it downward with no clamp; over ∼9 new levels a thermocline slope compounds into nonsense. Fixed by taking such water from the *nearest old node actually wet at that depth* — the horizontal donor the rest of the remap already uses. Water below the old seabed should look like the nearby ocean at that depth, not like a continuation of the local gradient.

Never hit before because every previous remap went mother → submesh: the new mesh was a strict *subset*, so no column ever got deeper. Ice *retreating* (208 810 → 209 172 nodes) exercises this path for the first time. (esm_tools `eef4f6db`.)

5 Where this leaves the strategy

1. **Look at PISM first.** A run whose shelves collapse 24%/decade is the anomaly. If PISM were equilibrated, the per-leg change would genuinely be small and the ocean side would very likely just work. **In particular: check the sub-shelf melt we are feeding PISM.** If our ocean hands PISM unrealistic melt, PISM's collapse is *our* doing — and that closes the loop.
2. **A per-leg $|\Delta \text{ulev}|$ cap** is worth having as a robustness guard (defer the extreme tail across legs, so the ocean never sees 16 levels open at once). But it papers over a PISM problem rather than fixing one.
3. **The fully-opened column** is now a narrow, well-defined case (donor water + zero velocity + η against still-cavity neighbours). Worth handling explicitly regardless.

6 Method note (continued)

The "mean $|\Delta\text{draft}| = 22\,\text{m}$" number was computed correctly and reported honestly — and was still deeply misleading, because a mean over a domain that is 98% unchanged says nothing about the tail that actually breaks the model. The distribution should have been plotted before it was used to justify a strategy. Two figures did in ten minutes what a summary statistic obscured for a day.